MIL-E-1/1158B
20 April 1973
SUPERSEDING
MIL-E-1/1158A
15 June 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 5736

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode

See figure 1

Mounting position: Vertical, anode up or down

Weight: 3.75-pounds nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter: / Unit:	Fl MHz	Ef Vac	Eb kVdc	Ec kVdc	Ib Adc	Ic Adc	Pp kW	Pi W
Maximum:	60	6.3	5.0	-1.0	1.4	0.5	2.5	5.0
Minimum:	--	5.7	---	----	---	---	---	---
TEST CONDITIONS:	--	6.0	2.5	Adjust	1.0	---	---	---

ABSOLUTE RATINGS: C Telegraphy

Parameter: / Unit:	Radiator temp °C (see note 1)	Glass temp °C (see note 2)
Maximum:	220	160
Minimum:	---	----
TEST CONDITIONS:	---	----

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
	Visual and mechanical inspection criteria		----	--	---	---	---	---
1301	Filament current		0.65	II	If	57	63	Aac
1261	Electrode voltage (grid)	Eb = 1,750 Vdc; Ec/Ib = 5 mAdc	0.65	II	Ec	-70	-125	Vdc
1231	Peak emission	eb = ec = 3,000 Vdc	0.65	II	is	30	---	a
1266	Total grid current	t = 120 seconds	0.65	II	Ic	---	-50	mAdc
1266	Primary grid emission	Ef = 6.0 Vac; Pg = 220 W; t = 60; anode grounded	0.65	II	Isg	---	30	mAdc
1261	Electrode voltage (1) (anode)	Ec = -30 Vdc; Eb/Ib = 0.4 Adc	0.65	II	Eb	1,400	2,020	Vdc
1261	Electrode voltage (2) (anode)	Ec = 0 Vdc; Eb/Ib = 0.4 Adc	0.65	II	Eb	850	1,250	Vdc
	Quality conformance inspection, part 2							
	Dimensions	See figure 1	6.5	S3	---	---	---	---
1316	Amplification factor	$\frac{Eb(1)-Eb(2)}{30}$	6.5	S3	Mu	18	26	---
1331	Direct interelectrode capacitance		6.5 6.5 6.5	S3 S3 S3	Cgp Cgk Cpk	12 15 0.2	21 22 1.0	pF pF pF
1236	Power oscillation	Eb = 5.0 kVdc; Ib = 1.0 Adc; Ic = 0.3 Adc; Rg = 2,670 ohms; F = 60 MHz	6.5	S3	Po	3.0	---	kW (useful)
1031	Vibration	No voltages	6.5	S3	---	---	---	---
1036	Bump	Angle = 15°	6.5	S3	---	---	---	---
	Quality conformance inspection, part 3							
	Life-test provisions	Group D; Power oscillation; t = 500 hours	---	---	---	---	---	---
	Life-test end points:							
1231	Peak emission		---	---	is	25	---	a
1266	Primary grid emission		---	---	Isg	---	30	mAdc
1236	Power oscillation		---	---	Po	2.6	---	kW

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NOTES:

1. An airflow of 150 cfm through the radiator must start with application of any voltages and may be discontinued with removal of all voltages. The inlet air temperature should not exceed 45°C. The pressure drop through the radiator shall not exceed 3 inches of water.

2. The glass may be cooled by deflecting a portion of the anode cooling air or by a separate blower delivering 10-cfm air through a 1-inch diameter nozzle.

Custodians:

Army - EL
Navy - EC
Air Force - 80

Review activities:

Army - EL
Navy - EC
Air Force - 17
DSA - ES

User activities:

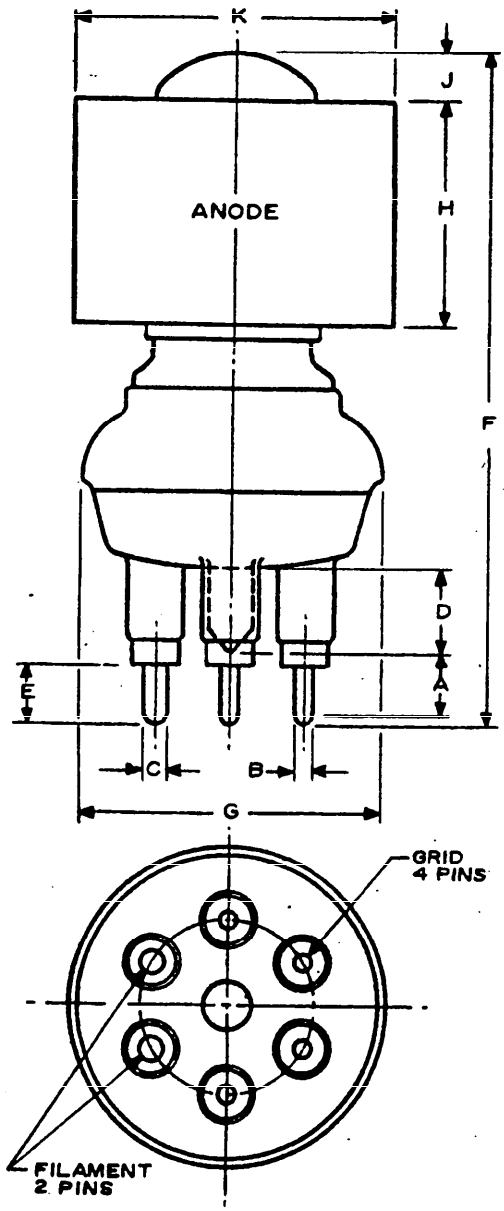
Army - MU
Navy - AS, CG, MC, OS, SH
Air Force - 11, 14

Preparing activity:

Air Force - 80

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
QUALITY CONFORMANCE INSPECT, PT.2		
A	.438 (11.13)	
B	.181 (4.60)	.195 (4.95)
C	.243 (6.17)	.257 (6.53)
D		1.000 (25.40)
E		.688 (17.48)
F		7.375 (187.33)
G		3.625 (92.08)
H	2.313 (58.75)	2.438 (61.93)
J		.563 (14.30)
K	3.438 (87.33)	3.563 (90.50)

NOTES:

a. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm

b. THE TUBE BASE PINS MUST SIMULTANEOUSLY ENTER TO A DISTANCE OF AT LEAST .500 (12.70 mm) INTO A FLAT GAGE HAVING 4 ADJACENT HOLES .288 ±.001 (7.32 ± .03 mm) DIA AND 2 ADJACENT HOLES .350 ±.001 (8.89 mm ± .03 mm) DIA, ALL LOCATED ON A 1.937 ± .001 (49.20 ± .03 mm) DIA BOLT CIRCLE, AT ANGLES OF 60° ± 10°. THERE SHALL BE A HOLE 1.000 ±.016 (25.4 mm ± .41 mm) IN THE GAGE CONCENTRIC WITH THE BOLT CIRCLE.

FIGURE 1. Outline drawing electron tube 5736

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